

Roberto André Mossi Milla

Software Engineer | Systems Infrastructure | Backend Development

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Summary

Software Engineer focused on backend systems, infrastructure, and data-intensive applications. Experience designing database-backed services, REST APIs, event-driven workflows, cloud storage integrations, containerized systems, and CI/CD pipelines across Rust, C++, Python, and TypeScript. Has built production systems spanning backend architecture, data processing, authentication, cloud services, and AI-enabled applications, with additional experience integrating computer vision and GPU-accelerated systems.

Experience

Ingeniería Digital

May 2026 – Present

Software Engineer (AI Systems)

Honduras

- Architected secure, database-backed services for healthcare document workflows using Next.js, TypeScript, PostgreSQL, Auth0, AWS S3, Amazon Textract, OpenAI APIs, and structured document generation.
- Designed document-ingestion workflows supporting file uploads, OCR extraction with Amazon Textract, cloud persistence, retrieval, version management, and user review of clinical records.
- Built data-acquisition services with Playwright to collect, normalize, and validate vehicle information from external sources for insurance workflows.
- Integrated AWS Rekognition for vehicle localization and Gemini Flash for image-based damage analysis within an AI-assisted insurance workflow, while keeping core business logic independent of model-specific implementations.
- Developed client and administrative applications backed by PostgreSQL and AWS S3, integrating AI and computer-vision services through structured application workflows and APIs.
- Established automated development and delivery workflows using GitHub Actions, Husky, ESLint, Prettier, Commitizen, and lint-staged across production repositories.

NOIR

Jan. 2026 – Present

Founder & Software Engineer (In Development)

Remote

- Architecting a Rust backend using Axum, PostgreSQL, Redis, authentication, and cloud services for a cross-platform movie discovery application.
- Designing backend services for metadata aggregation, availability tracking, synchronization, and data access across native Apple applications.
- Owning system architecture, data modeling, API design, deployment planning, automated testing, internationalization, and integration across backend and client applications.

Gannon University

Aug. 2024 – May 2025

Computer Vision Engineer — Industry-Sponsored Project

Erie, PA

- Built components of an industrial traceability platform using Dockerized services, event-driven processing, TimescaleDB, and operational dashboards for real-time warehouse monitoring.
- Integrated Ultralytics YOLOv8 computer-vision events with persistent time-series storage and monitoring interfaces, connecting image-processing pipelines with backend services and operational data.
- Developed dataset-management and model-retraining workflows, including interfaces for correcting YOLOv8 detection data and persisting updated annotations.

Gannon University

Aug. 2024 – May 2025

Research Assistant — Evolutionary Artificial Intelligence

Erie, PA

- Developed GPU-accelerated multi-agent simulations in Unreal Engine using C++, CUDA, and PyTorch, with performance-conscious processing and experimentation workflows.
- Implemented the NEAT evolutionary algorithm and genome representation in C++, developing procedural simulation environments for autonomous agents.
- Co-authored peer-reviewed research published through the American Society for Engineering Education (ASEE) and presented results at Sigma Xi.

Dinant

Jun. 2023 – Jul. 2023

Full-Stack Developer

Honduras

- Developed production data-backed workflow systems using Oracle SQL, JavaScript, HTML, and CSS, contributing to database design, reporting, filtering, and business-process automation.
- Built and deployed applications that replaced manual workflows requiring weeks with automated processes completed in minutes.
- Worked directly with stakeholders to translate operational requirements into database-driven applications and production software.

Technical Skills

Backend & Systems: Rust, Axum, C++, Python, Node.js, REST APIs, PostgreSQL, Redis, TimescaleDB, Oracle Database, SQL, event-driven services

Infrastructure & Cloud: Docker, Linux, AWS S3, GitHub Actions, CI/CD, CMake, Git, cloud storage, authentication

AI & Computer Vision: OpenAI APIs, Gemini Flash, AWS Rekognition, Amazon Textract, Ultralytics YOLOv8, PyTorch, CUDA, NEAT, OCR, multimodal AI

Languages: Rust, C++, Python, TypeScript, JavaScript, SQL, Java, Bash, Lua
Data & Processing: Database design, data pipelines, document processing, image processing, computer vision, Playwright
Engineering Tools: Husky, ESLint, Prettier, Commitizen, lint-staged, automated testing, deployment workflows

Education & Recognition

Gannon University	2025
<i>B.S. Computer Systems Engineering</i>	<i>Erie, PA</i>
• Peer-reviewed publication with the American Society for Engineering Education (ASEE); research presentation at Sigma Xi.	